Notes on Spectral (Fourier) Method of Integration

PHYS4840 / Joyce / Classs 26

Overview

The **spectral method** is a numerical technique for solving differential equations by expanding functions in terms of global basis functions. When using a **Fourier basis**, these basis functions are sines and cosines (or complex exponentials), and the function is expressed as a Fourier series:

$$f(x) \approx \sum_{k=-N}^{N} \hat{f}_k e^{ikx}$$

Here, $\hat{f}_k$ are the Fourier coefficients, and e^{ikx} are the basis functions.

Integration Using the Fourier Spectral Method

Integration in the spectral method becomes algebraic. If a function is represented by its Fourier series:

$$f(x) = \sum_{k=-N}^{N} \hat{f}_k e^{ikx}$$

then its indefinite integral is:

$$F(x) = \sum_{k \neq 0} \frac{\hat{f}_k}{ik} e^{ikx} + C$$

The term for $k = 0$ corresponds to a constant in $f(x)$, and its integral becomes a linear function, which must be handled separately. The constant C is the integration constant.

Example: Integrating $f(x) = \cos(x)$

We consider the function

$$f(x) = \cos(x)$$

on the domain $[-\pi, \pi]$.

Step 1: Fourier Series

The cosine function can be written using Euler's formula:

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

So the nonzero Fourier coefficients are:

$$\hat{f}_1 = \hat{f}_{-1} = \frac{1}{2}$$

Step 2: Integrate Term-by-Term

We integrate each mode separately:

$$F(x) = \sum_{k \neq 0} \frac{\hat{f}_k}{ik} e^{ikx} = \frac{1}{2i} \left(\frac{e^{ix}}{1} - \frac{e^{-ix}}{1} \right) = \frac{1}{2i} (e^{ix} - e^{-ix}) = \sin(x)$$

So the result of integrating $\cos(x)$ is:

$$\int \cos(x)\, dx = \sin(x) + C$$

which matches the standard result.

Practical Implementation with FFTs

In practical numerical applications:

1. Discretize $f(x)$ on a uniform grid.
2. Compute the FFT to obtain $\hat{f}_k$.
3. Integrate in Fourier space by dividing by ik (excluding $k = 0$).
4. Apply the inverse FFT to get $F(x)$.

This approach is highly accurate for smooth periodic functions and forms the basis of spectral numerical methods used in computational physics and fluid dynamics.